Grade 12 Mathematics

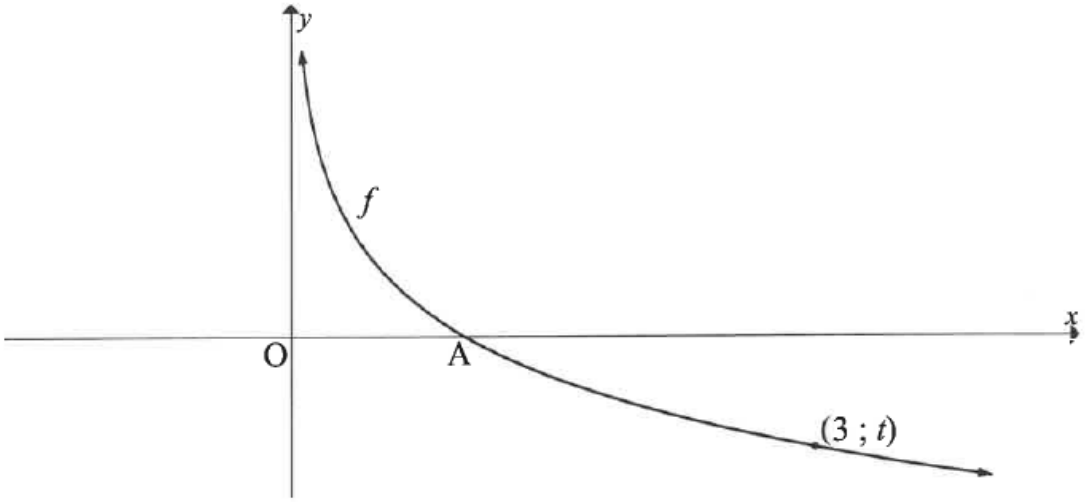
Functions & Inverses

Revision Questions

NOV 2025

QUESTION 4

The graph of $f(x)=\log_{\frac{1}{3}} x$ is drawn below. Point A is the x-intercept of f and $(3\ ;\ t)$ lies on f.



4.1 Calculate the value of t (1)

4.2 Write down the coordinates of A. (1)

4.3 Determine the equation of f^{-1}, the inverse of f, in the form $y=\ldots$ (2)

4.4 Write down the equation of the asymptote of f^{-1} (1)

4.5 Draw the graph of f^{-1} on the set of axes provided in the ANSWER BOOK. Clearly indicate the intercepts with the axes, the coordinates of ONE other point and the asymptotes. (3)

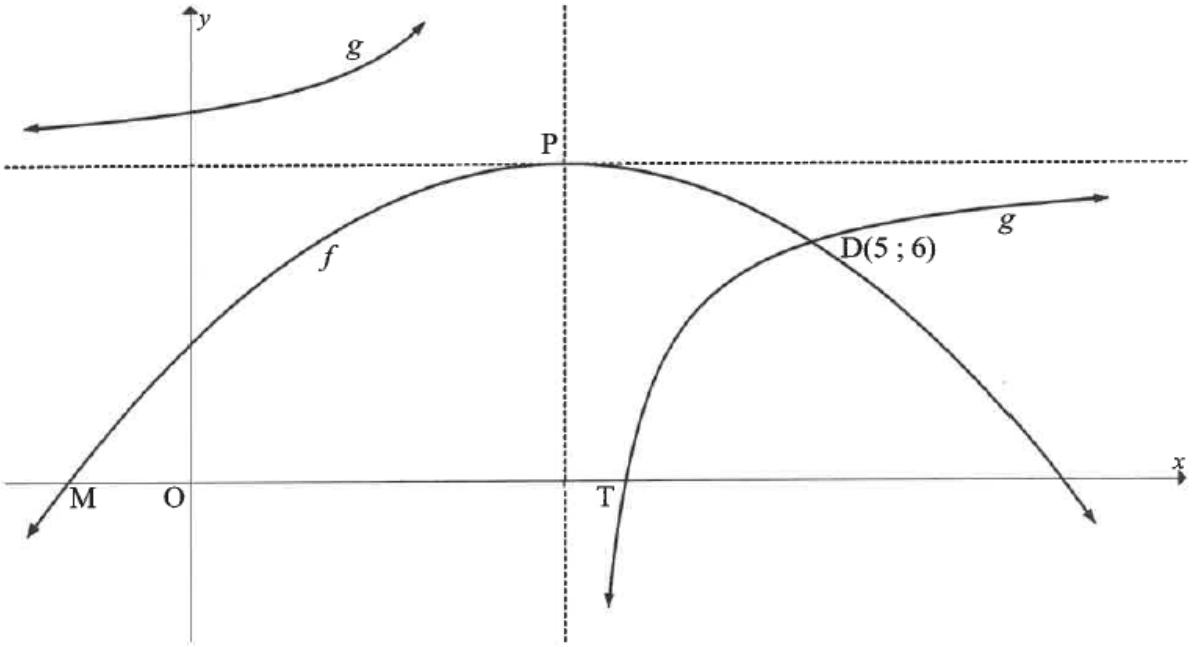
4.6 The graph of h is obtained when f^{-1} is translated 5 units to the right. Determine the y-values of h where $x>4$. (2)

[10]

QUESTION 5

The graphs of $f(x)=ax^2+bx+c$ and $g(x)=\frac{-4}{x-3}+8$ are drawn below.

P is the turning point of f and the point of intersection of the asymptotes of g.
The graphs intersect at D(5 ; 6). M and T are x-intercepts of f and g respectively.



5.1 Write down the domain of g. (1)

5.2 Write down the range of f. (1)

5.3 Determine the values of x for which:

5.3.1 $g(x) \leq f(x)$ (2)

5.3.2 $f(x) < 6$ (2)

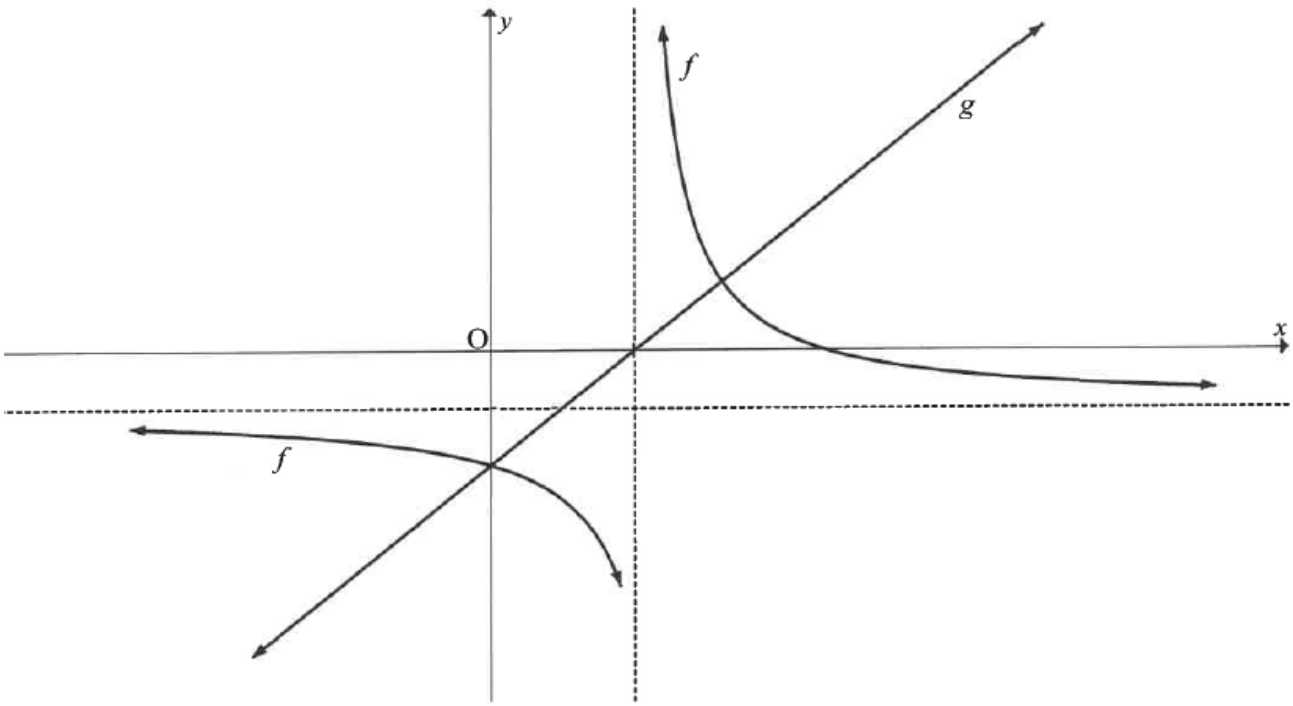
5.4 Show that the equation of the parabola is $f(x)=-\frac{1}{2}x^2+3x+\frac{7}{2}$ (3)

5.5 Calculate the length of MT. (6)

5.6 Determine the equation of the tangent to f at D. (3)
[18]

QUESTION 6

The graphs of $g(x) = x + c$ and $f(x) = \frac{a}{x+p} + q$ are drawn below. Graph g and the vertical asymptote of f intersect at the x-axis.



6.1 Write down the coordinates of the x-intercept of g in terms of p. (1)

6.2 Graph g intersects the horizontal asymptote of graph f at $x = 1$ and the graph f at $x = 3$. Graphs f and g also intersect on the y-axis. Determine the equation of f. (5)

6.3 Describe the transformation that g must undergo to become an axis of symmetry of f that cuts f at two points. (2)
[8]

QUESTION 4

Given: $f(x) = a^x - 1$ for $a > 0$. B$\left(2; \frac{-5}{9}\right)$ is a point on f.

4.1 Calculate the value of a. (2)

4.2 Write down the range of f. (1)

4.3 Sketch the graph of f. Clearly show the intercepts with the axes and asymptotes, if any. (3)

4.4 It is further given that C is a point on f at $y = \frac{19}{8}$.

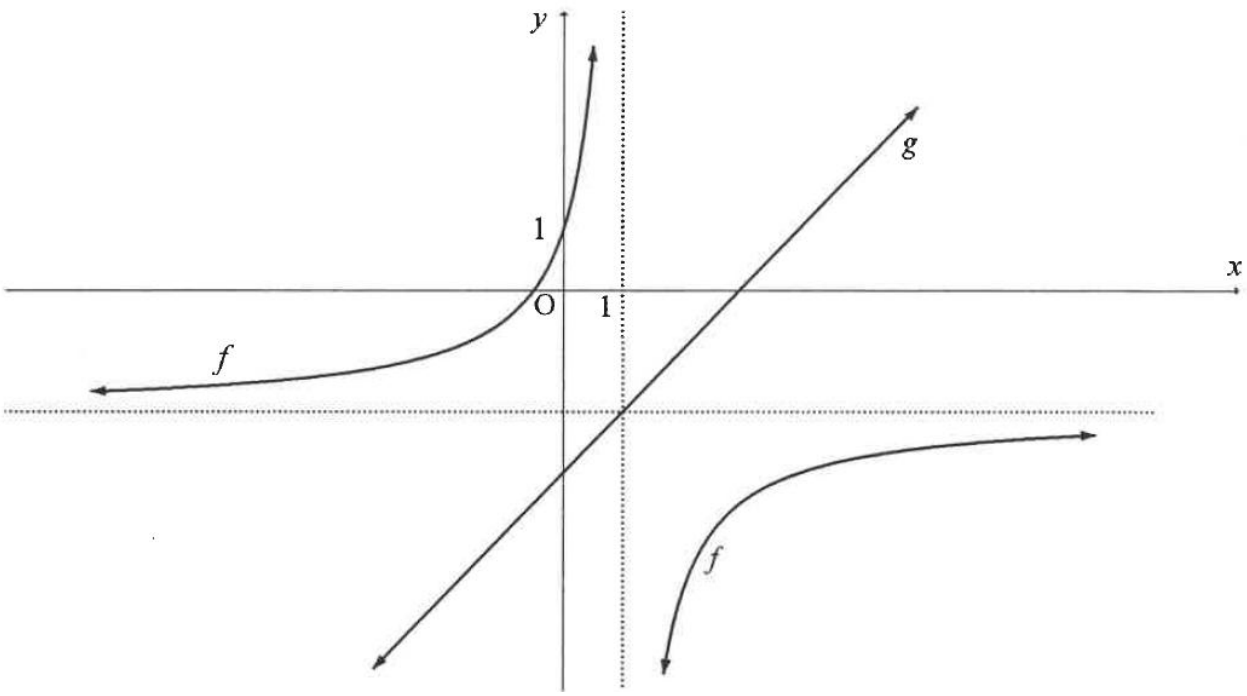
Determine the coordinates of C$'$, the image of C, when C is reflected about the line $y = x$. (3)

[9]

QUESTION 5

Sketched below is the graph of $f(x) = \frac{a}{x+p} + q$ having the domain $(-\infty; 1) \cup (1; \infty)$.

The graph of f cuts the y-axis at $(0; 1)$. A line of symmetry of f is given by $g(x) = x - 3$.



5.1 Write down the value of p. (1)

5.2 Determine the equation of the horizontal asymptote of f. (2)

5.3 Calculate the value of a. (2)

5.4 For which values of x is $f(x) \geq 0$? (3)

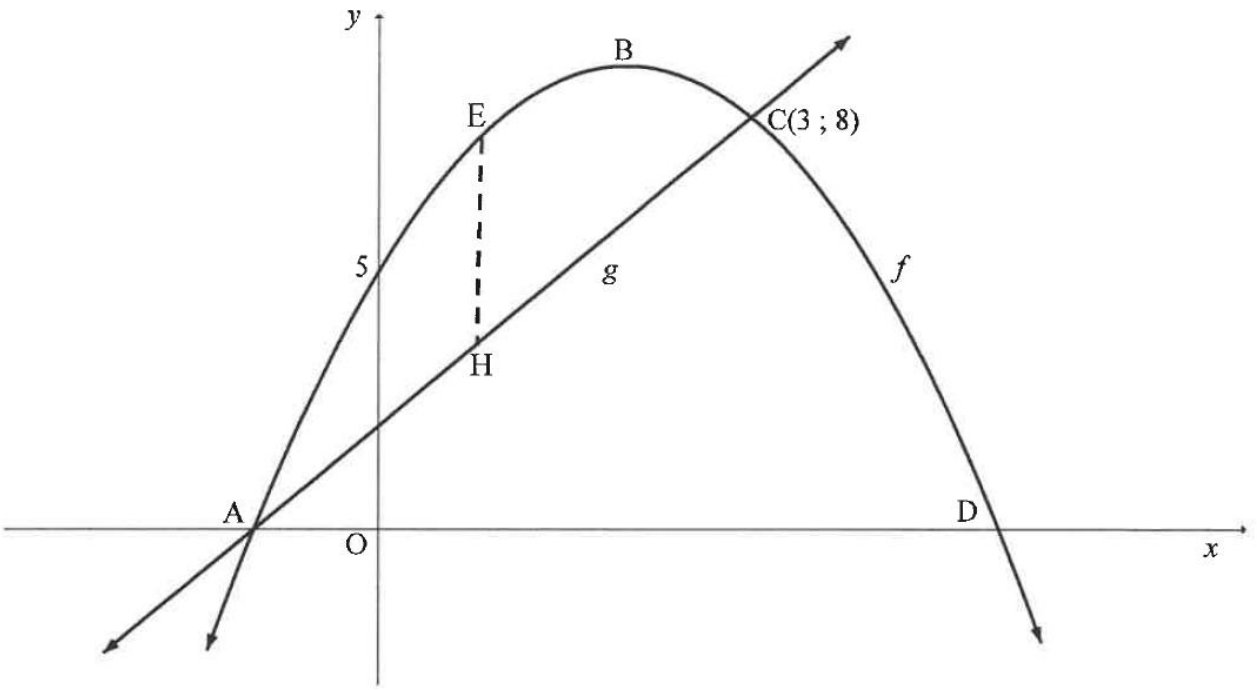
5.5 Graph f undergoes a transformation to h where:

- The domain and range of h are the same as that of f
- $h'(x)$, the derivative of h, is negative on its domain

Describe a possible transformation that f could have undergone to result in h. (2)
[10]

QUESTION 6

In the diagram below, the graphs of $f(x) = -x^2 + 4x + 5$ and g, a straight line, are drawn. $C(3\,;8)$ is a point of intersection of f and g. EH is drawn parallel to the y-axis, with E a point on f and H a point on g.



6.1 Calculate the coordinates of B, the turning point of f. (3)

6.2 Show that the equation of the line through A and C is given by $g(x) = 2x + 2$. (3)

6.3 Calculate the maximum length of EH for $f > g$. (4)

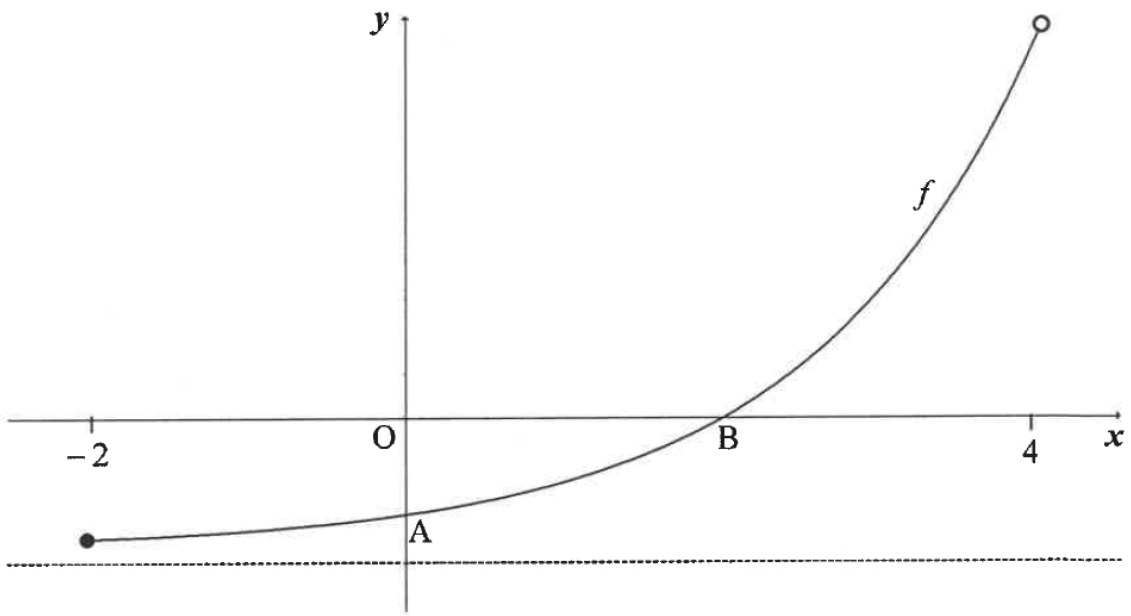
6.4 Given: $k(x) = f(x+m) = -x^2 - 2mx - m^2 + 4x + 4m + 5$

Determine the value of m such that g is a tangent to k. (5)
[15]

NOV 2023

QUESTION 4

Sketched below is the graph of $f(x) = 2^x - 4$ for $x \in [-2; 4)$.
A and B are respectively the y- and x-intercepts of f.



4.1 Write down the equation of the asymptote of f. (1)

4.2 Determine the coordinates of B. (2)

4.3 Determine the equation of k, a straight line passing through A and B in the form $k(x) = \ldots$ (3)

4.4 Calculate the vertical distance between k and f at $x = 1$ (3)

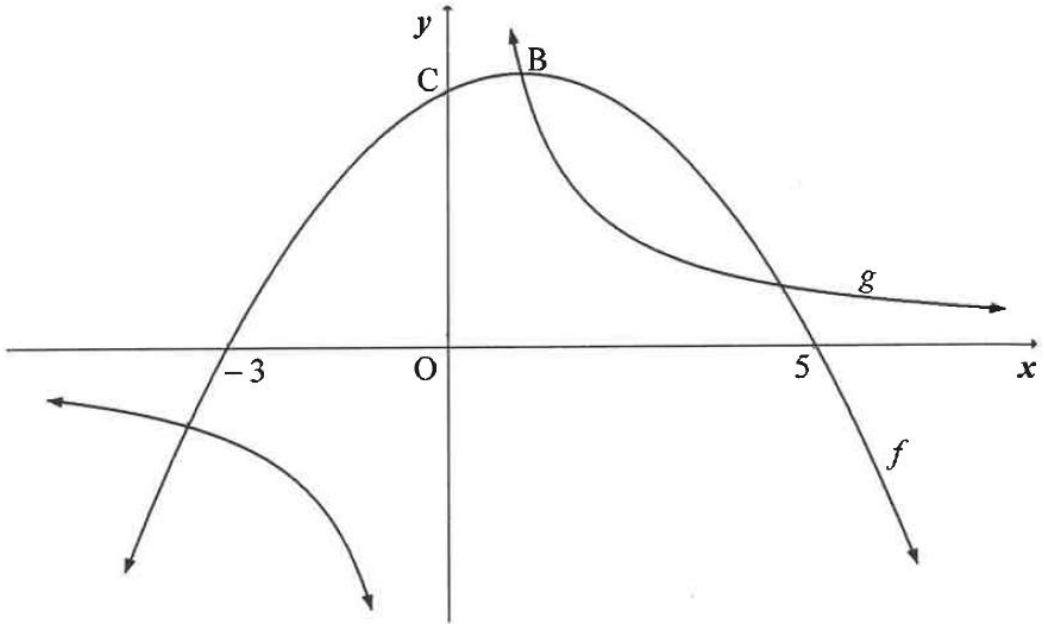
4.5 Write down the equation of g if it is given that $g(x) = f(x) + 4$ (1)

4.6 Write down the domain of g^{-1}. (2)

4.7 Write down the equation of g^{-1} in the form $y = \ldots$ (2)
[14]

QUESTION 5

The graphs of $f(x) = -\frac{1}{2}(x-1)^2 + 8$ and $g(x) = \frac{d}{x}$ are drawn below. A point of intersection of f and g is B, the turning point of f. The graph f has x-intercepts at $(-3\ ;\ 0)$ and $(5\ ;\ 0)$ and a y-intercept at C.



5.1 Write down the coordinates of the turning point of f. (2)

5.2 Calculate the coordinates of C. (2)

5.3 Calculate the value of d. (1)

5.4 Write down the range of g. (1)

5.5 For which values of x will $f(x).g(x) \leq 0$? (3)

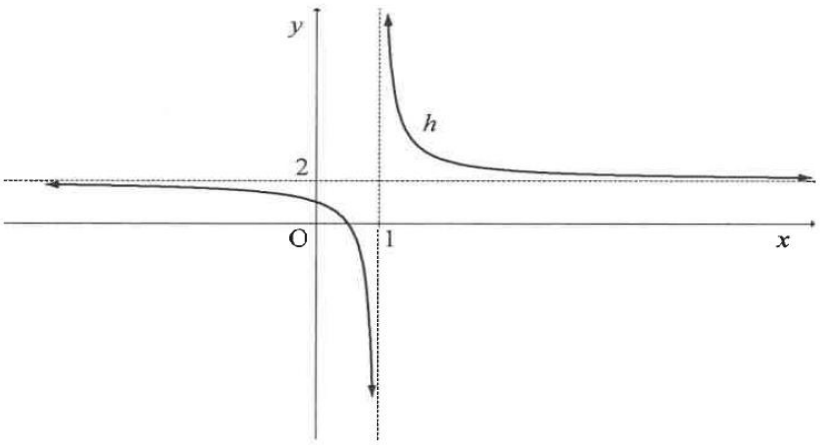
5.6 Calculate the values of k so that $h(x) = -2x + k$ will not intersect the graph of g. (5)

5.7 h is a tangent to g at R, a point in the first quadrant. Calculate t such that $y = f(x) + t$ intersects g at R. (4)

[18]

QUESTION 4

4.1 Sketched below is the graph of $h(x)=\frac{1}{x+p}+q$. The asymptotes of h intersect at $(1\ ;2)$.



4.1.1 Write down the values of p and q. (2)

4.1.2 Calculate the coordinates of the x-intercept of h. (2)

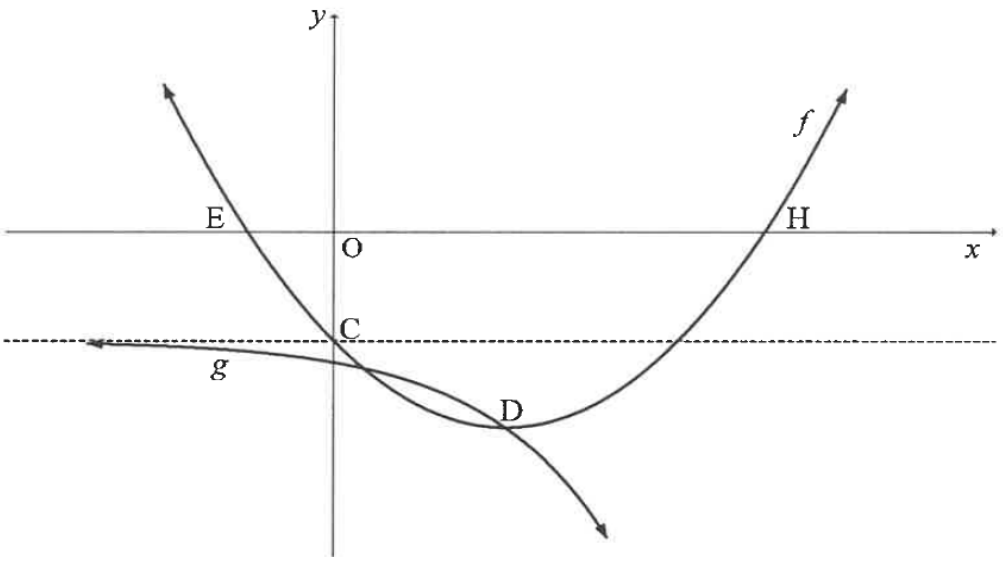
4.1.3 Write down the x-coordinate of the x-intercept of g if $g(x)=h(x+3)$. (2)

4.1.4 The equation of an axis of symmetry of h is $y=x+t$. Determine the value of t. (2)

4.1.5 Determine the values of x for which $-2\leq\frac{1}{x-1}$. (3)

4.2 The graphs of $f(x)=x^2-4x-5$ and $g(x)=a.2^x+q$ are sketched below.

- E and H are the x-intercepts of f.
- C is the y-intercept of f and lies on the asymptote of g.
- The two graphs intersect at D, the turning point of f.



4.2.1 Write down the y-coordinate of C. (1)

4.2.2 Determine the coordinates of D. (2)

4.2.3 Determine the values of a and q. (3)

4.2.4 Write down the range of g. (1)

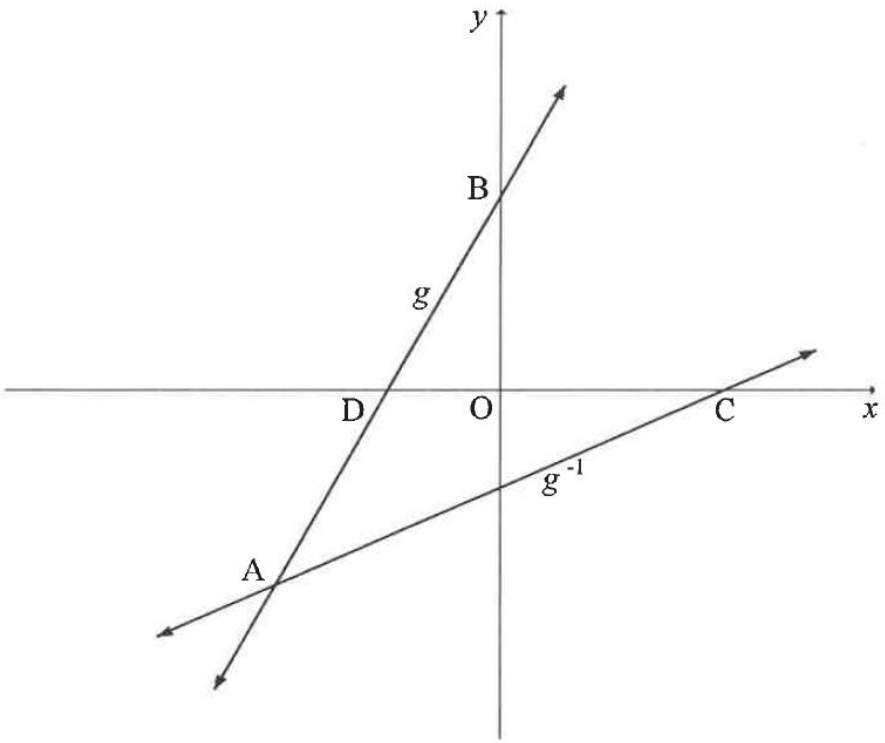
4.2.5 Determine the values of k for which the value of $f(x) - k$ will always be positive. (2)

[20]

QUESTION 5

The graphs of $g(x) = 2x + 6$ and g^{-1}, the inverse of g, are shown in the diagram below.

- D and B are the x- and y-intercepts respectively of g.
- C is the x-intercept of g^{-1}.
- The graphs of g and g^{-1} intersect at A.



5.1 Write down the y-coordinate of B. (1)

5.2 Determine the equation of g^{-1} in the form $g^{-1}(x) = mx + n$. (2)

5.3 Determine the coordinates of A. (3)

5.4 Calculate the length of AB. (2)

5.5 Calculate the area of ΔABC. (5)

[13]

QUESTION 5

Given: $f(x)=\frac{-1}{x-3}+2$

5.1 Write down the equations of the asymptotes of f. (2)

5.2 Write down the domain of f. (1)

5.3 Determine the coordinates of the x-intercept of f. (2)

5.4 Write down the coordinates of the y-intercept of f. (2)

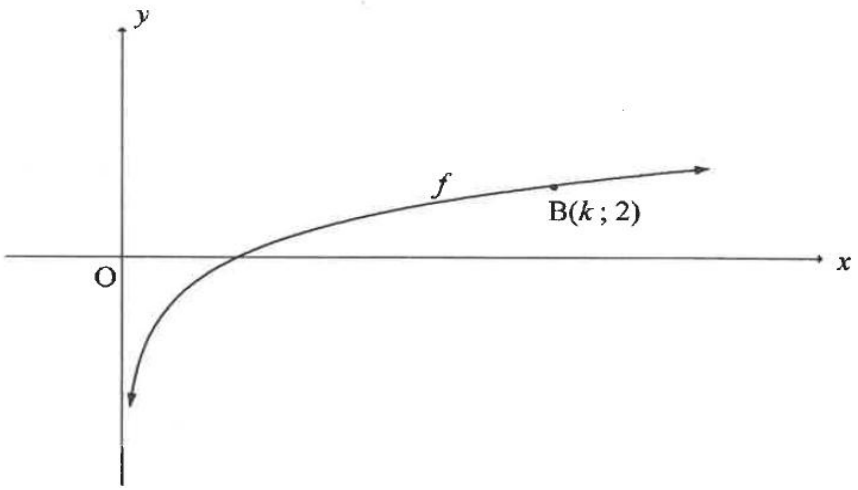
5.5 Draw the graph of f. Clearly show ALL the asymptotes and intercepts with the axes. (3)

[10]

QUESTION 6

The graph of $f(x)=\log_4 x$ is drawn below.

B$(k\,;2)$ is a point on f.



6.1 Calculate the value of k. (2)

6.2 Determine the values of x for which $-1 \le f(x) \le 2$. (2)

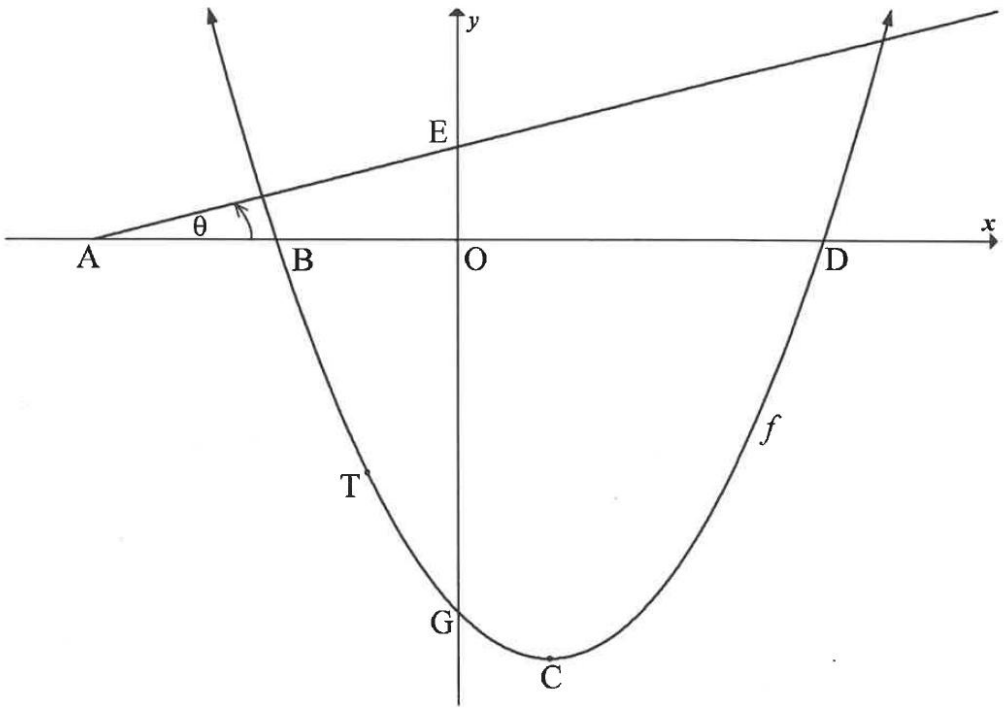
6.3 Write down the equation of f^{-1}, the inverse of f, in the form $y=\ldots$ (2)

6.4 For which values of x will $x.f^{-1}(x)<0$? (2)

[8]

QUESTION 7

The graph of $f(x) = (x+4)(x-6)$ is drawn below.
The parabola cuts the x-axis at B and D and the y-axis at G.
C is the turning point of f.
Line AE has an angle of inclination of θ and cuts the x-axis and y-axis at A and E respectively.
T is a point on f between B and G.



7.1 Write down the coordinates of B and D. (2)

7.2 Calculate the coordinates of C. (2)

7.3 Write down the range of f. (1)

7.4 Given that $\theta = 14{,}04°$ and the tangent to f at T is perpendicular to AE.

7.4.1 Calculate the gradient of AE, correct to TWO decimal places. (1)

7.4.2 Calculate the coordinates of T. (5)

7.5 A straight line, g, parallel to AE, cuts f at K(–3 ; –9) and R.
Calculate the x-coordinate of R. (6)
[17]

QUESTION 4

4.1 Given: $h(x)=\frac{-3}{x-1}+2$

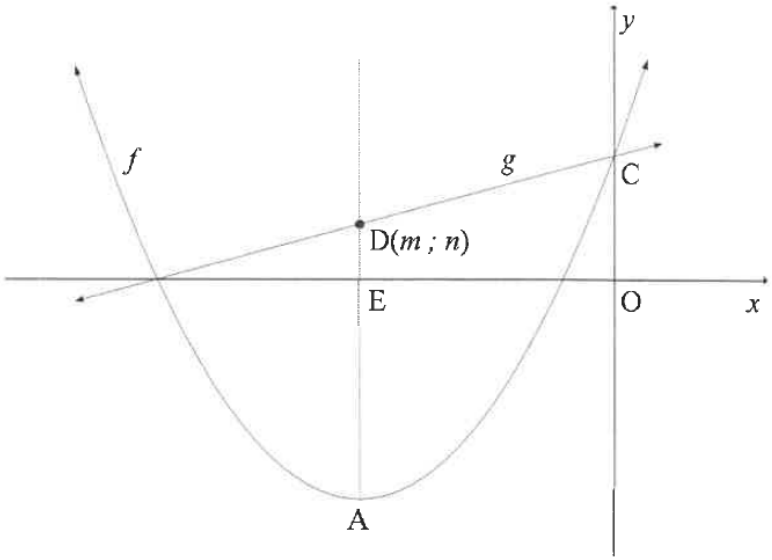
4.1.1 Write down the equations of the asymptotes of h. (2)

4.1.2 Determine the equation of the axis of symmetry of h that has a negative gradient. (2)

4.1.3 Sketch the graph of h, showing the asymptotes and the intercepts with the axes. (4)

4.2 The graphs of $f(x)=\frac{1}{2}(x+5)^2-8$ and $g(x)=\frac{1}{2}x+\frac{9}{2}$ are sketched below.

- A is the turning point of f.
- The axis of symmetry of f intersects the x-axis at E and the line g at D(m ; n).
- C is the y-intercept of f and g.



4.2.1 Write down the coordinates of A. (2)

4.2.2 Write down the range of f. (1)

4.2.3 Calculate the values of m and n. (3)

4.2.4 Calculate the area of OCDE. (3)

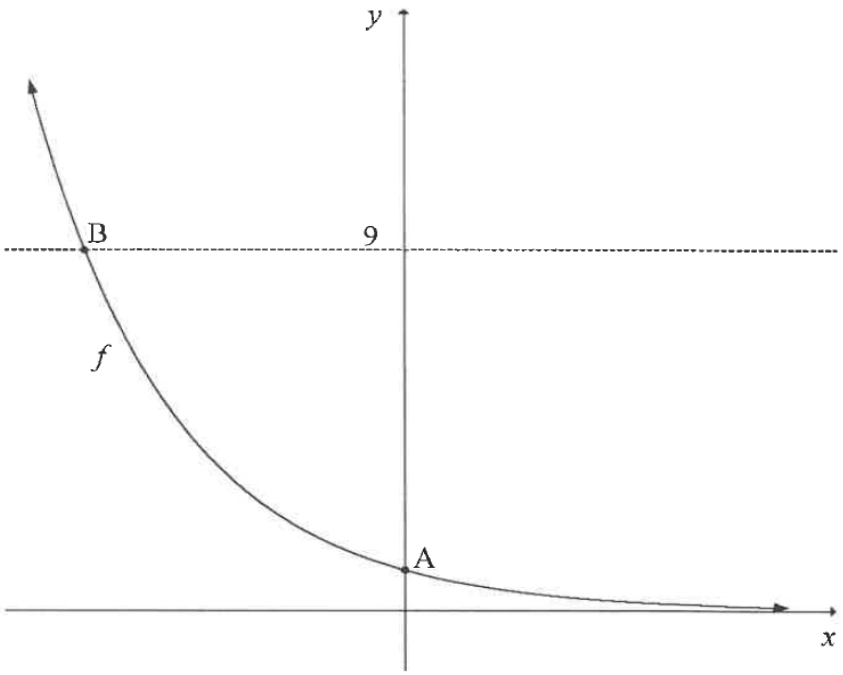
4.2.5 Determine the equation of g^{-1}, the inverse of g, in the form $y=\ldots$ (2)

4.2.6 If $h(x)=g^{-1}(x)+k$ is a tangent to f, determine the coordinates of the point of contact between h and f. (4)

[23]

QUESTION 5

The graph of $f(x)=3^{-x}$ is sketched below. A is the y-intercept of f.
B is the point of intersection of f and the line $y=9$.



5.1 Write down the coordinates of A. (1)

5.2 Determine the coordinates of B. (3)

5.3 Write down the domain of f^{-1}. (2)

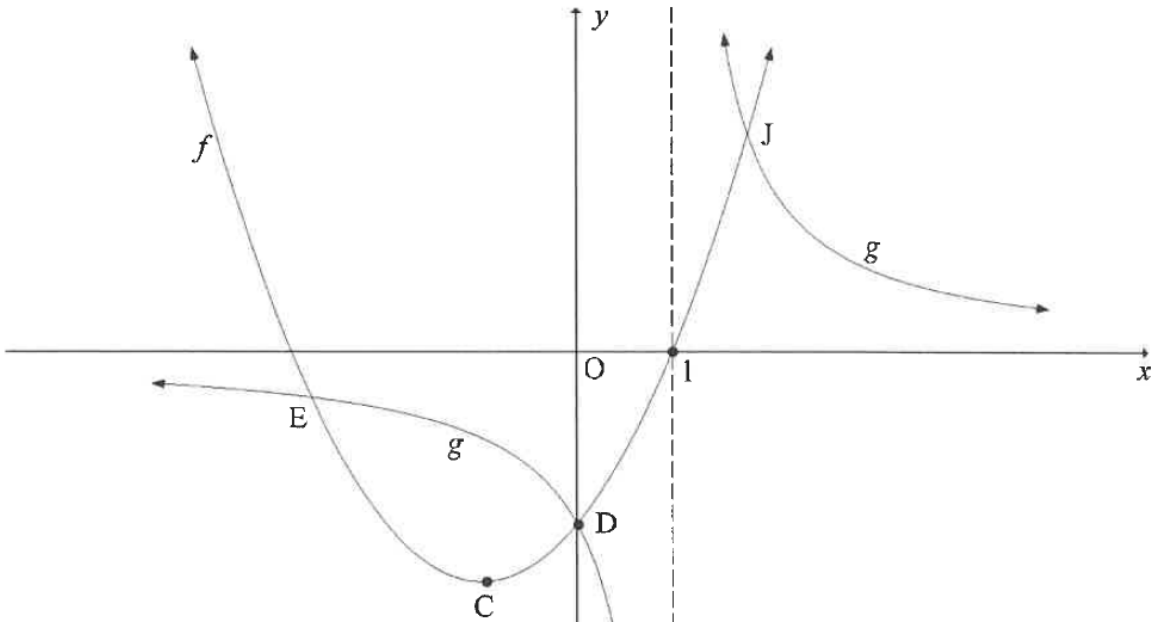
5.4 Describe the translation from f to $h(x)=\frac{27}{3^x}$. (3)

5.5 Determine the values of x for which $h(x)<1$. (3)
[12]

QUESTION 4

Below are the graphs of $f(x) = x^2 + bx - 3$ and $g(x) = \dfrac{a}{x+p}$.

- f has a turning point at C and passes through the x-axis at $(1\ ;\ 0)$.
- D is the y-intercept of both f and g. The graphs f and g also intersect each other at E and J.
- The vertical asymptote of g passes through the x-intercept of f.



4.1 Write down the value of p. (1)

4.2 Show that $a = 3$ and $b = 2$. (3)

4.3 Calculate the coordinates of C. (4)

4.4 Write down the range of f. (2)

4.5 Determine the equation of the line through C that makes an angle of $45°$ with the positive x-axis. Write your answer in the form $y = \ldots$ (3)

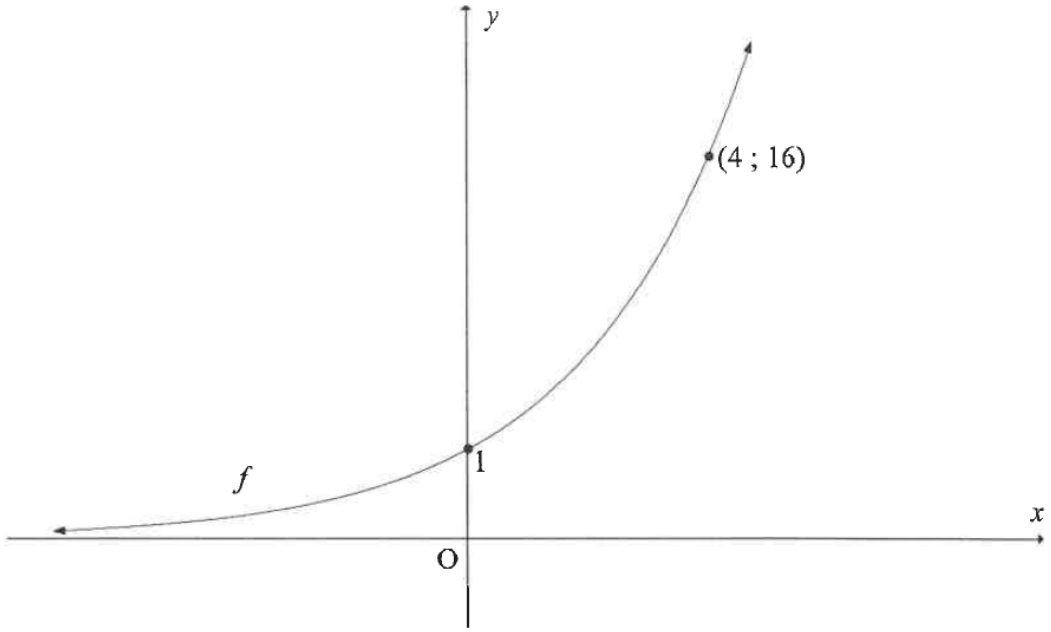
4.6 Is the straight line, determined in QUESTION 4.5, a tangent to f? Explain your answer. (2)

4.7 The function $h(x) = f(m - x) + q$ has only one x-intercept at $x = 0$. Determine the values of m and q. (4)

[19]

QUESTION 5

Sketched below is the graph of $f(x) = k^x$; $k > 0$. The point $(4 ; 16)$ lies on f.



5.1 Determine the value of k. (2)

5.2 Graph g is obtained by reflecting graph f about the line $y = x$. Determine the equation of g in the form $y = \ldots$ (2)

5.3 Sketch the graph g. Indicate on your graph the coordinates of two points on g. (4)

5.4 Use your graph to determine the value(s) of x for which:

5.4.1 $f(x) \times g(x) > 0$ (2)

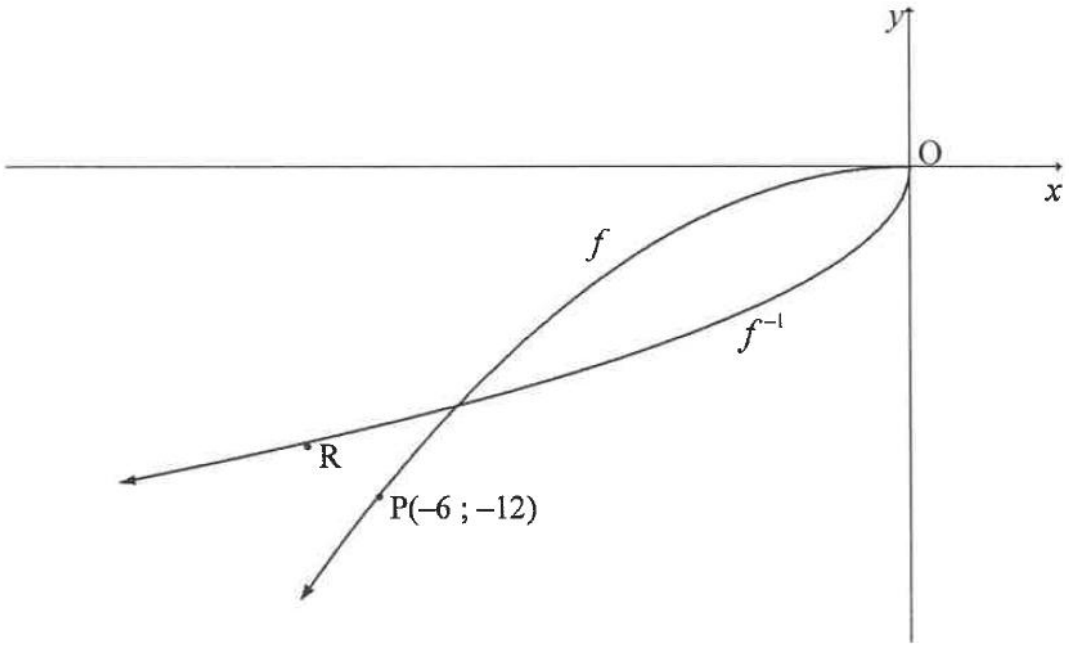
5.4.2 $g(x) \leq -1$ (2)

5.5 If $h(x) = f(-x)$, calculate the value of x for which $f(x) - h(x) = \frac{15}{4}$ (4)

[16]

QUESTION 4

In the diagram below, the graph of $f(x) = ax^2$ is drawn in the interval $x \leq 0$.
The graph of f^{-1} is also drawn. P(–6 ; –12) is a point on f and R is a point on f^{-1}.



4.1 Is f^{-1} a function? Motivate your answer. (2)

4.2 If R is the reflection of P in the line $y = x$, write down the coordinates of R. (1)

4.3 Calculate the value of a. (2)

4.4 Write down the equation of f^{-1} in the form $y = ...$ (3)
[8]

QUESTION 5

Given: $f(x) = \dfrac{-1}{x-1}$

5.1 Write down the domain of f. (1)

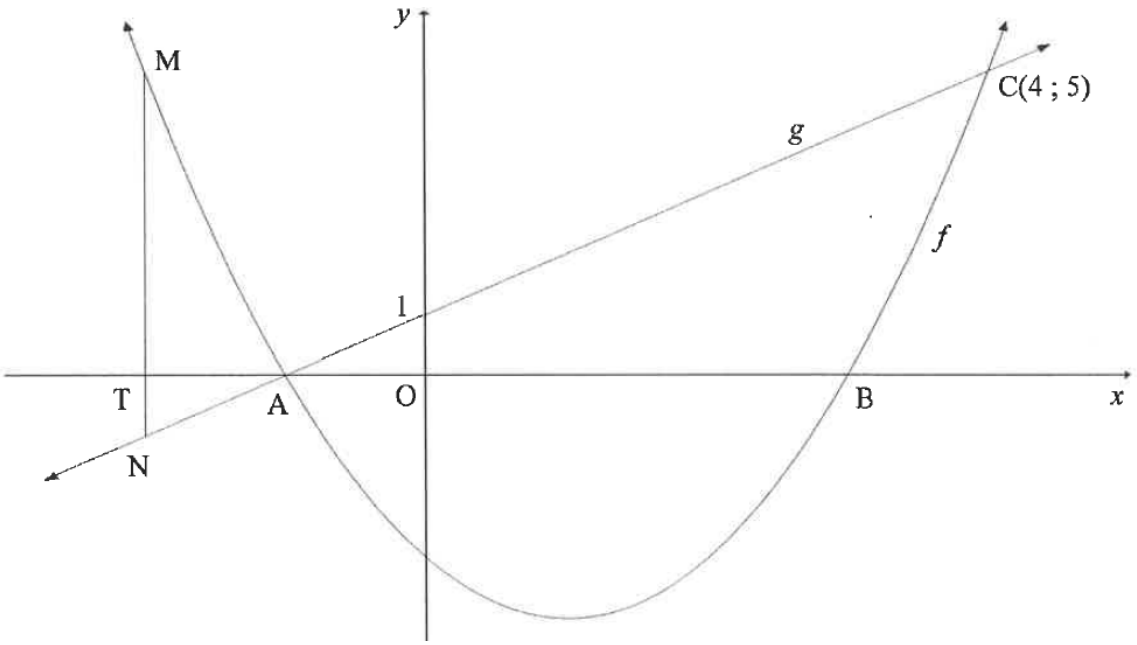
5.2 Write down the asymptotes of f. (2)

5.3 Sketch the graph of f, clearly showing all intercepts with the axes and any asymptotes. (3)

5.4 For which values of x will $x.f'(x) \geq 0$? (2)
[8]

QUESTION 6

In the diagram below, A and B are the x-intercepts of the graph of $f(x) = x^2 - 2x - 3$.
A straight line, g, through A cuts f at C(4 ; 5) and the y-axis at (0 ; 1).
M is a point on f and N is a point on g such that MN is parallel to the y-axis.
MN cuts the x-axis at T.



6.1 Show that $g(x) = x + 1$. (2)

6.2 Calculate the coordinates of A and B. (3)

6.3 Determine the range of f. (3)

6.4 If MN = 6:

6.4.1 Determine the length of OT if T lies on the negative x-axis. Show ALL your working. (4)

6.4.2 Hence, write down the coordinates of N. (2)

6.5 Determine the equation of the tangent to f drawn parallel to g. (5)

6.6 For which value(s) of k will $f(x) = x^2 - 2x - 3$ and $h(x) = x + k$ NOT intersect? (1)
[20]

QUESTION 4

Given: $f(x) = -ax^2 + bx + 6$

4.1 The gradient of the tangent to the graph of f at the point $\left(-1\ ;\ \frac{7}{2}\right)$ is 3.

Show that $a = \frac{1}{2}$ and $b = 2$. (5)

4.2 Calculate the x-intercepts of f. (3)

4.3 Calculate the coordinates of the turning point of f. (3)

4.4 Sketch the graph of f. Clearly indicate ALL intercepts with the axes and the turning point. (4)

4.5 Use the graph to determine the values of x for which $f(x) > 6$. (3)

4.6 Sketch the graph of $g(x) = -x - 1$ on the same set of axes as f. Clearly indicate ALL intercepts with the axes. (2)

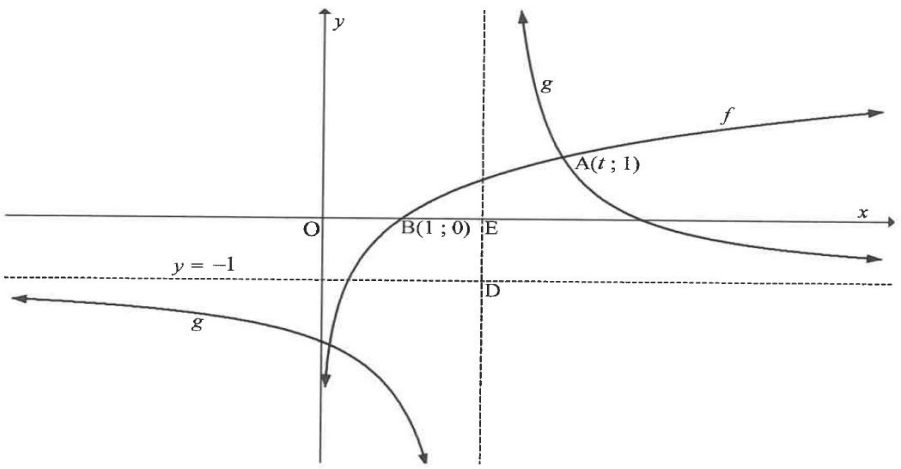
4.7 Write down the values of x for which $f(x).g(x) \leq 0$. (3)

[23]

QUESTION 5

The diagram below shows the graphs of $g(x) = \frac{2}{x+p} + q$ and $f(x) = \log_3 x$.

- $y = -1$ is the horizontal asymptote of g.
- B(1 ; 0) is the x-intercept of f.
- A(t ; 1) is a point of intersection between f and g.
- The vertical asymptote of g intersects the x-axis at E and the horizontal asymptote at D.
- OB = BE.



5.1 Write down the range of g. (2)

5.2 Determine the equation of g. (2)

5.3 Calculate the value of t. (3)

5.4 Write down the equation of f^{-1}, the inverse of f, in the form $y=\ldots$ (2)

5.5 For which values of x will $f^{-1}(x)<3$? (2)

5.6 Determine the point of intersection of the graphs of f and the axis of symmetry of g that has a negative gradient. (3)

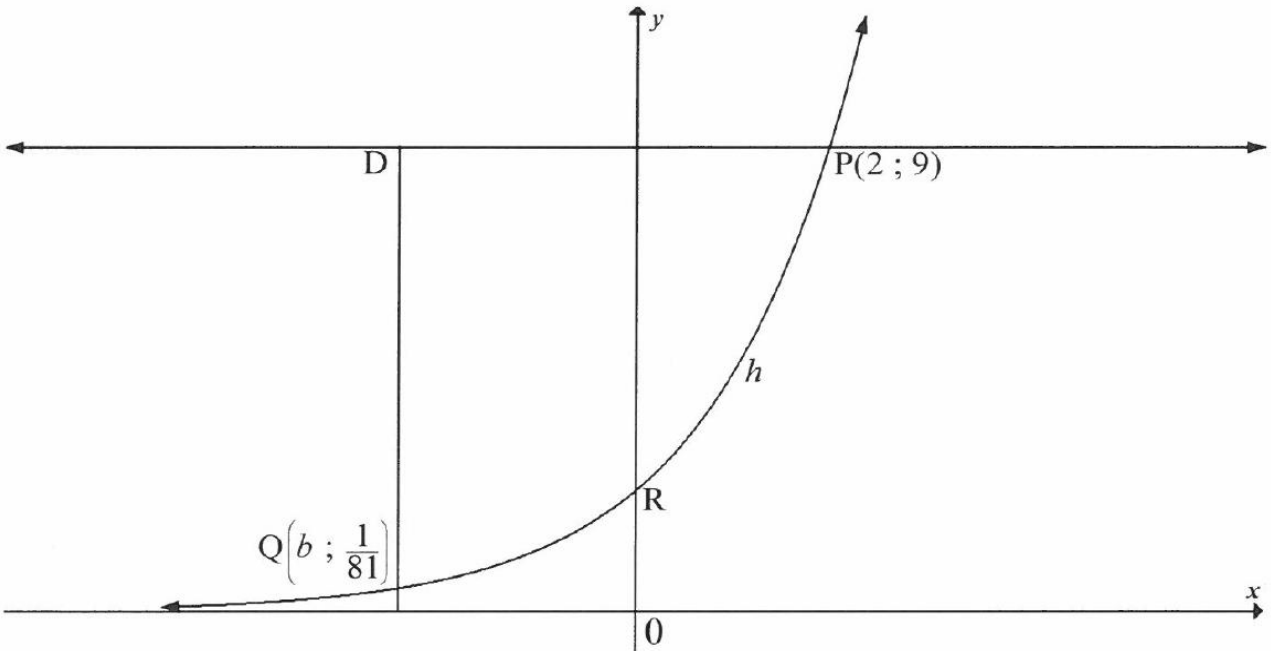
[14]

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QUESTION 4

Sketched below is the graph of $h(x)=a^x$, $a>0$. R is the y-intercept of h.

The points P(2 ; 9) and $Q\left(b\,;\frac{1}{81}\right)$ lie on h.



4.1 Write down the equation of the asymptote of h. (1)

4.2 Determine the coordinates of R. (1)

4.3 Calculate the value of a. (2)

4.4 D is a point such that DQ $\|$ y-axis and DP $\|$ x-axis. Calculate the length of DP. (4)

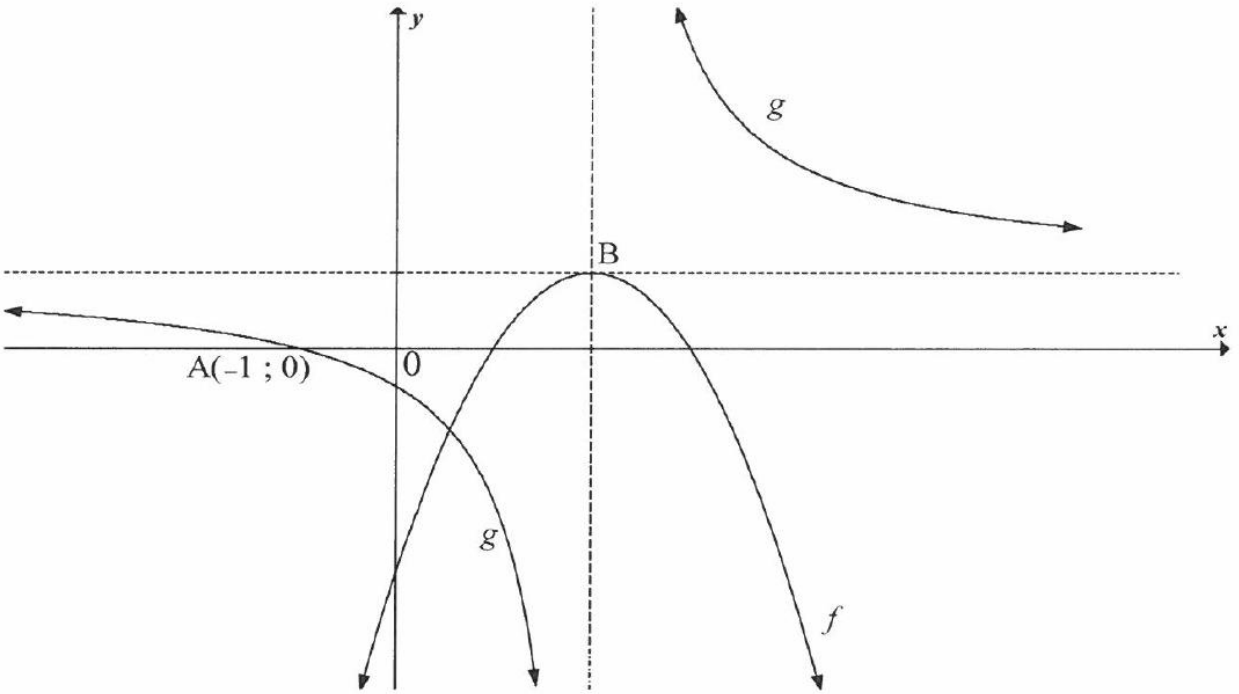
4.5 Determine the values of k for which the equation $h(x+2)+k=0$ will have a root that is less than -6. (3)

[11]

QUESTION 5

Sketched below is the parabola f, with equation $f(x) = -x^2 + 4x - 3$ and a hyperbola g, with equation $(x - p)(y + t) = 3$.

- B, the turning point of f, lies at the point of intersection of the asymptotes of g.
- A$(-1\,;\,0)$ is the x-intercept of g.



5.1 Show that the coordinates of B are $(2\,;\,1)$ (2)

5.2 Write down the range of f. (1)

5.3 For which value(s) of x will $g(x) \geq 0$? (2)

5.4 Determine the equation of the vertical asymptote of the graph of h if $h(x) = g(x + 4)$ (1)

5.5 Determine the values of p and t. (4)

5.6 Write down the values of x for which $f(x).g'(x) \geq 0$ (4)

[14]

QUESTION 6

Given: $f(x) = -x + 3$ and $g(x) = \log_2 x$

6.1 On the same set of axes, sketch the graphs of f and g, clearly showing ALL intercepts with the axes. (4)

6.2 Write down the equation of $g^{-1}(x)$, the inverse of g, in the form $y = \ldots$ (2)

6.3 Explain how you will use QUESTION 6.1 and/or QUESTION 6.2 to solve the equation $\log_2(3-x) = x$. (3)

6.4 Write down the solution to $\log_2(3-x) = x$. (1)
[10]

NOV 2015

QUESTION 4

Given: $f(x) = 2^{x+1} - 8$

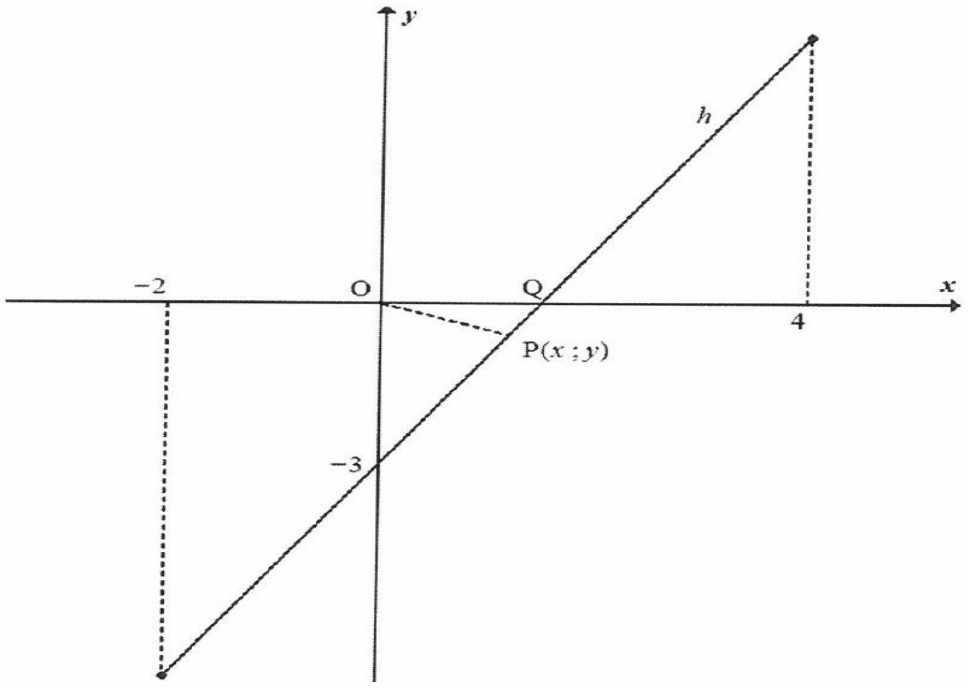
4.1 Write down the equation of the asymptote of f. (1)

4.2 Sketch the graph of f. Clearly indicate ALL intercepts with the axes as well as the asymptote. (4)

4.3 The graph of g is obtained by reflecting the graph of f in the y-axis. Write down the equation of g. (1)
[6]

QUESTION 5

Given: $h(x) = 2x - 3$ for $-2 \le x \le 4$. The x-intercept of h is Q.



5.1 Determine the coordinates of Q. (2)

5.2 Write down the domain of h^{-1}. (3)

5.3 Sketch the graph of h^{-1} in your ANSWER BOOK, clearly indicating the y-intercept and the end points. (3)

5.4 For which value(s) of x will $h(x) = h^{-1}(x)$? (3)

5.5 P$(x\ ;\ y)$ is the point on the graph of h that is closest to the origin. Calculate the distance OP. (5)

5.6 Given: $h(x) = f'(x)$ where f is a function defined for $-2 \le x \le 4$.

5.6.1 Explain why f has a local minimum. (2)

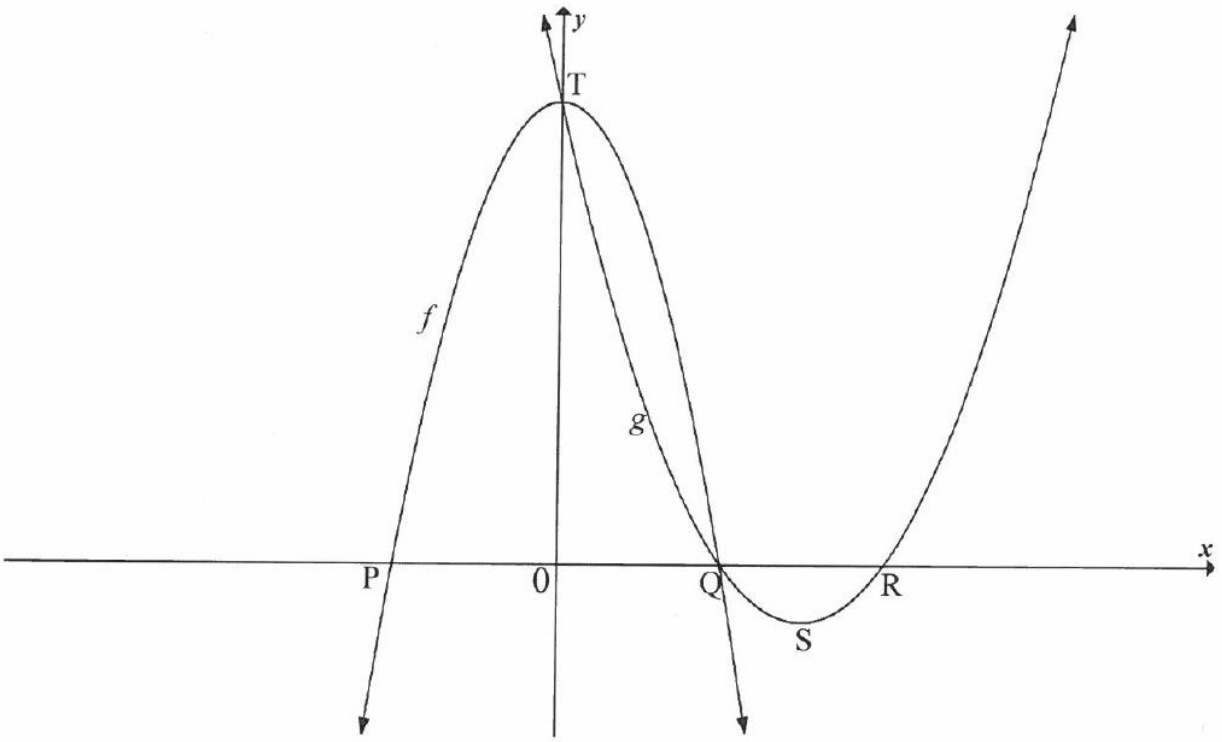
5.6.2 Write down the value of the maximum gradient of the tangent to the graph of f. (1)

[19]

QUESTION 6

6.1 The graphs of $f(x) = -2x^2 + 18$ and $g(x) = ax^2 + bx + c$ are sketched below.

Points P and Q are the x-intercepts of f. Points Q and R are the x-intercepts of g. S is the turning point of g. T is the y-intercept of both f and g.



6.1.1 Write down the coordinates of T. (1)

6.1.2 Determine the coordinates of Q. (3)

6.1.3 Given that $x = 4{,}5$ at S, determine the coordinates of R. (2)

6.1.4 Determine the value(s) of x for which $g''(x) > 0$. (2)

6.2 The function defined as $y = \frac{a}{x+p} + q$ has the following properties:

- The domain is $x \in R,\ x \neq -2$.
- $y = x + 6$ is an axis of symmetry.
- The function is increasing for all $x \in R,\ x \neq -2$.

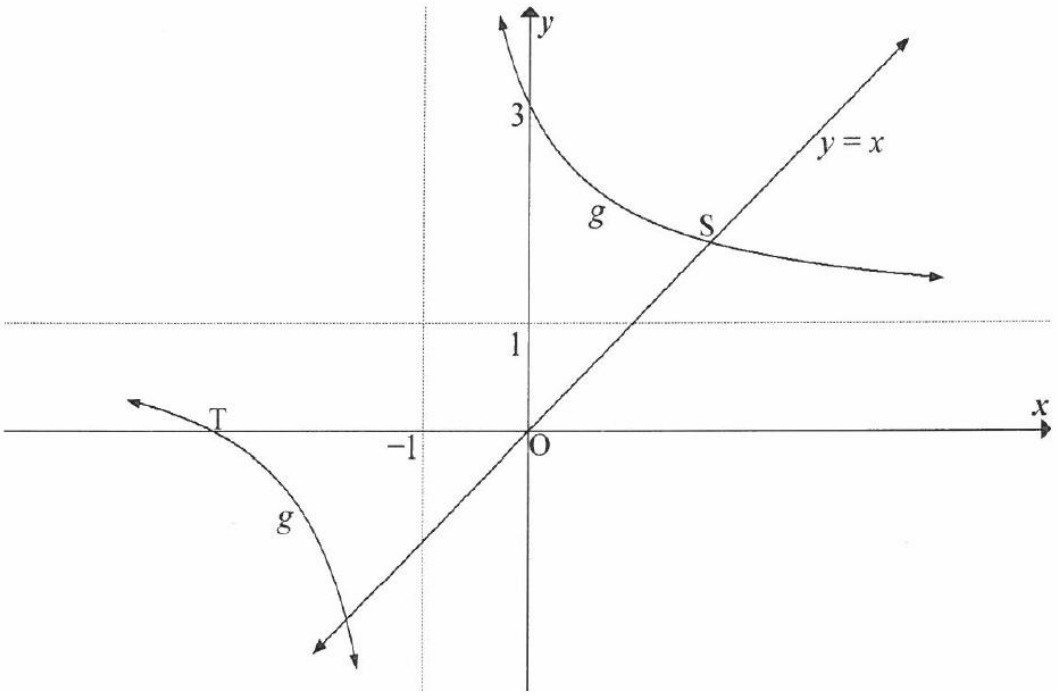
Draw a neat sketch graph of this function. Your sketch must include the asymptotes, if any. (4)

[12]

NOV 2014

QUESTION 4

The diagram below shows the hyperbola g defined by $g(x) = \frac{2}{x+p} + q$ with asymptotes $y = 1$ and $x = -1$. The graph of g intersects the x-axis at T and the y-axis at $(0\ ;\ 3)$. The line $y = x$ intersects the hyperbola in the first quadrant at S.



4.1 Write down the values of p and q. (2)

4.2 Calculate the x-coordinate of T. (2)

4.3 Write down the equation of the vertical asymptote of the graph of h, if $h(x) = g(x+5)$ (1)

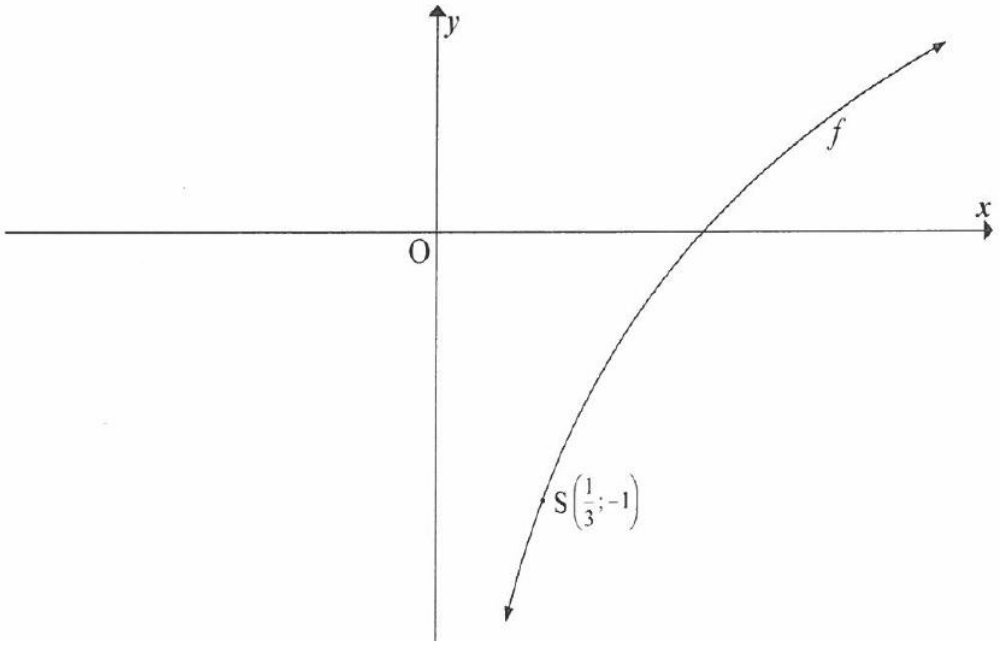
4.4 Calculate the length of OS. (5)

4.5 For which values of k will the equation $g(x) = x + k$ have two real roots that are of opposite signs? (1)

[11]

QUESTION 5

Given: $f(x) = \log_a x$ where $a > 0$. $S\left(\frac{1}{3};-1\right)$ is a point on the graph of f.



5.1 Prove that $a = 3$. (2)

5.2 Write down the equation of h, the inverse of f, in the form $y = ...$ (2)

5.3 If $g(x) = -f(x)$, determine the equation of g. (1)

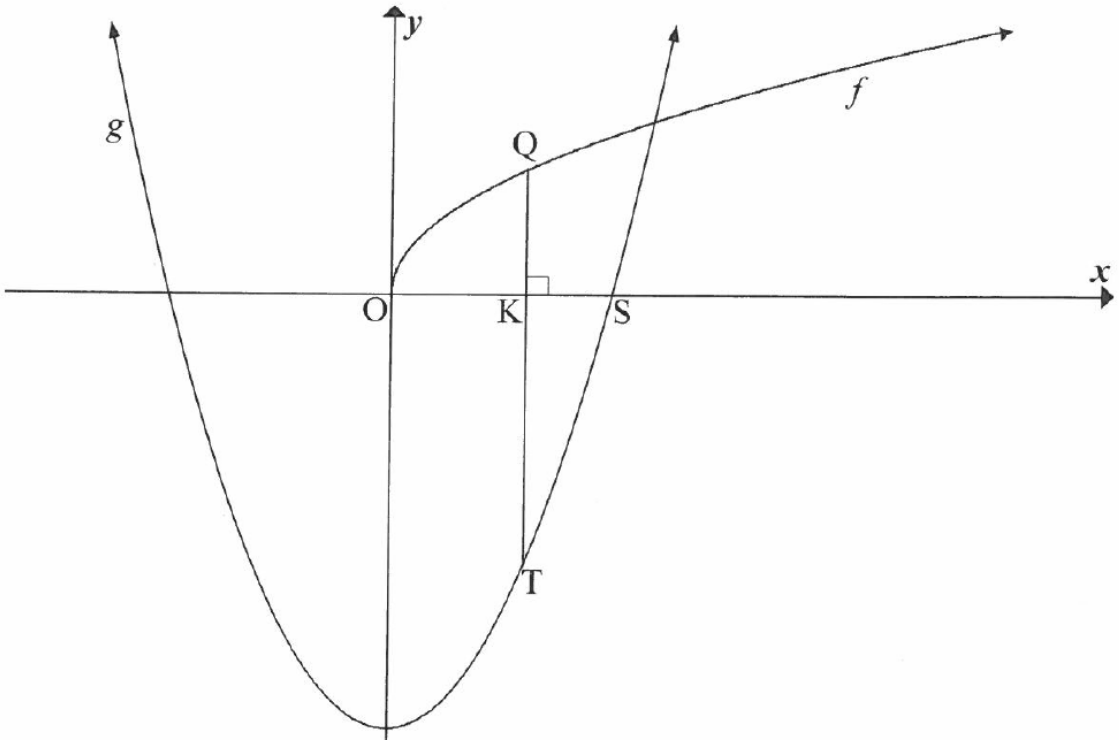
5.4 Write down the domain of g. (1)

5.5 Determine the values of x for which $f(x) \geq -3$. (3)

[9]

QUESTION 6

Given: $g(x) = 4x^2 - 6$ and $f(x) = 2\sqrt{x}$. The graphs of g and f are sketched below. S is an x-intercept of g and K is a point between O and S. The straight line QKT with Q on the graph of f and T on the graph of g, is parallel to the y-axis.



6.1 Determine the x-coordinate of S, correct to TWO decimal places. (2)

6.2 Write down the coordinates of the turning point of g. (2)

6.3 6.3.1 Write down the length of QKT in terms of x, where x is the x-coordinate of K. (3)

6.3.2 Calculate the maximum length of QT. (6)
[13]

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